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KHWOPA COLLEGE OF ENGINEERING

AFFILIATED TO TRIBHUVAN UNIVERSITY

A
Report on

LAB 3: FIXED CAPACITOR THYRISTOR CONTROLLED REACTOR

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LAB 3

FIXED CAPACITOR THYRISTOR CONTROLLED REACTOR

1 Objectives

To understand the working of a fixed capacitor thyristor controlled reactor(FCTCR) and analyze the waveforms in MATLAB/Simulink.

2 Software Used

- MATLAB/Simulink.

3 Theory

A basic VAR generator using a fixed capacitor with a thyristor controlled reactor is as shown in the Figure 1(a) below:

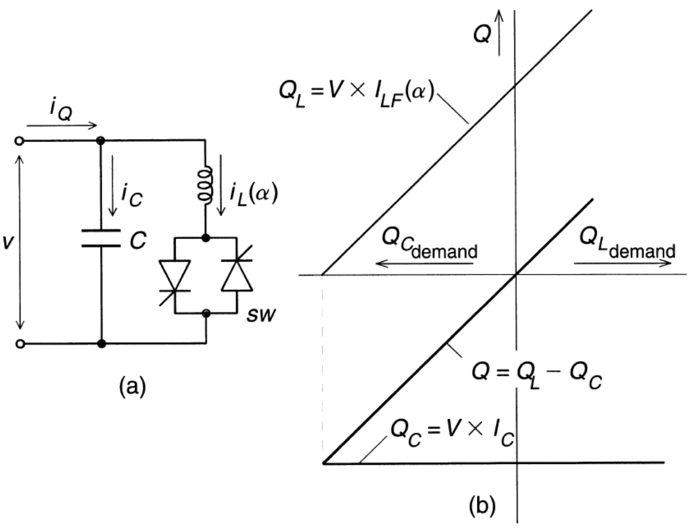

Figure 1: Fixed Capacitor Thyristor Controlled Reactor

The current in the reactor is varied similarly to the thyristor controlled reactor method. The FCTCR may be considered essentially to consist of a variable reactor (controlled by a delay angle α) and a fixed capacitor. The constant capacitive var generation of the fixed capacitor is opposed by the variable var absorption (Q_l) of the thyristor controlled reactor to yeild the total var output required as seen in Figure 1(b). At zero var output, the capacitive and inductive currents become equal and thus the capacitive and inductive vars cancel out. With a further decrease of angle α the inductive current becomes larger than the capacitive current, resulting in a net inductive var output. At zero var output , the capacitive and inductive currents become equal and thus the capacitve and inductive vars cancel out. With a further decrease of angle α the inductive current becomes larger than the capacitive current resulting in a net inductive var output. At zero delay angle, the thyristor controlled reactor conducts current over the full $180°$ interval, resulting in maximum inductive var output that is equal to the var generated by the capacitor and those absorbed by the fully conducting reactor.

When the inductive load decreases, the fixed capacitor wil overcompensate. This over-compensation is neutralized reducing the firing angle of tcr branch with the help of a closed loop control system.